

Ushtrime ND9e

2.3.1. Të gjendet shtypja e $n=1$ mol gaz i cili në temperaturën $T=300\text{ K}$ zë vëllimin $V=1\text{ l}$?

Zgjidhja:

Të dhënat:

$$n=1\text{ mol}$$

$$T=300\text{ K}$$

$$V=1\text{ l}=1\cdot 10^{-3}\text{ m}^3$$

$$p=?$$

$$R=8,134 \frac{\text{J}}{\text{mol}\cdot\text{K}}$$

$$v_m^2 = \langle v^2 \rangle$$

$$p = \frac{1}{3} n_0 m \langle v^2 \rangle$$

$$E_k = \frac{1}{2} m \langle v^2 \rangle / \cdot 2$$

$$2E_k = m \langle v^2 \rangle$$

$$\underline{E_k = E_{tr}}$$

$$E_{tr} = \frac{3}{2} \frac{R}{N_A} \cdot T$$

$$p = \frac{2}{3} n_0 E_k$$

$$p = \frac{2}{3} n_0 \frac{3}{2} \cdot \frac{R}{N_A} T$$

$$p = n_0 \frac{R}{N_A} \cdot T$$

$$\boxed{n_0 = \frac{N}{V}}$$

$$p = \frac{N}{N_A} \frac{RT}{V}$$

$$\boxed{n = \frac{N}{N_A}}$$

$$p = \frac{nRT}{V}$$

$$\boxed{p \cdot V = nRT} \text{ ek. i Klapayronit}$$

$$p = \frac{nRT}{V} = \frac{1\text{ mol} \cdot 8,134 \frac{\text{J}}{\text{mol}\cdot\text{K}} \cdot 300\text{ K}}{1 \cdot 10^{-3}\text{ m}^3} = 2493 \cdot 10^3 \text{ Pa}$$

$$p = 24,93 \cdot 10^5 \text{ Pa} = 24,93 \underline{\underline{\text{bar}}}$$

$$\boxed{1 \cdot 10^5 \text{ Pa} = 1 \text{ bar}}$$

$$\frac{\text{J}}{\text{m}^3} = \frac{\text{N} \cdot \text{m}}{\text{m}^3} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$$